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University of Essex
MSc Artificial Intelligence
Module: IA_PCOM7E January 2026
Unit 12: Assessment – Individual Reflection

Individual Reflection

I. Introduction

The module coursework focused on two assignments: a group design project for an automated, agent-driven software system, followed by an individual code implementation of that same system.

The module overall, along with both assignments, are reviewed using the heuristic framework of the Gibbs' Reflective Cycle (“the cycle”), a standard approach for personal reflection on the learning process (Praveena et al, 2025). Summarized, the cycle follows six steps: Description, Feelings, Evaluation, Analysis, Conclusion, and Action Plan (Praveena et al, 2025).

Accordingly, each step of the cycle is discussed in order:

II. Description

The Description step of the Gibbs' Reflective Cycle summarizes the course of learning undertaken (Praveena et al, 2025). For this module’s assignments:

1. The group design project focused on developing a technical specification, a design proposal brief, and several architectural diagrams for an automated content moderation system for user-generated images uploaded to a generic social media platform (Appendix A; Bullen, 2026). For the group project, aside from general discussion within the group, I wrote the entirety of the design proposal brief and designed the diagram included here in Appendix A.
2. The individual development project focused on implementing, with approval from my tutor in advance, an extract of the group design proposal into a fully functional live demo (Appendix B; Bullen, 2026). My implementation focused on the core functionality of the system: categorising images by the content type contained in them using the AWS Rekognition platform, then applying evaluative business rules to those AI-determined categories either to allow

posting of the image, to deny it, or to mark the image for further review by human moderators.

III. Feelings

The Feelings step of the Gibbs' Reflective Cycle examines subjective motivation to engage with the subject matter of the work undertaken (Praveena et al, 2025). Overall, I felt that that the module and both assignments were effective ways to gain a deeper familiarity with agentic design patterns in a practical, real world context.

IV. Evaluation & Analysis

The Evaluation step reviews which parts of the work garnered successful outcomes, versus those that presented learning challenges or other issues (Praveena et al, 2025). Tightly couple to the Evaluation step, the Analysis step of the Gibbs' Reflective Cycle examines lessons learned and identifies opportunities for future improvement (Praveena et al, 2025).

In terms of learning outcomes, I thought that the module was successful in providing a thorough overview of the current state of the art for agentic system development. In terms of learning challenges and coursework challenges, there were several, albeit not insurmountable ones:

For the group design project, coordinating work schedules across time zone differences was challenging, but resolved by using the module’s group project forum and one of the group member’s private Discord server for asynchronous discussion.

For the individual development project, I found it both interesting and challenging to reduce the original architectural diagram – which was highly complex, since it included over a dozen software agents, along with supporting components such as inter-agent message queues, databases, and user interfaces – down to a manageable scope that still reflected the entirety of the system’s core image categorization functionality.

I was able to accomplish this objective in a fully functional web-based demo. In hindsight, it did also mean that I needed to leave out a few optional components that were simply too complex for one person to build in the allotted time. For example, an optional, but interesting, component would have been to shuttle the results of human moderation of images that the AI image categorization platform could not reliably identify (or apply evaluative business rules to) into a neural net for the automated development of new evaluative business rules that the system could incorporate with minimal human intervention. To do so to an objectively thorough degree of development, however, would have required several thousand human moderation decisions made on a corpus of at least several dozen training images, as well as building out a custom neural net and the attendant back end web infrastructure, which made implementing such a component unfeasible.

I also found it worthwhile, but challenging, to build out the agent components using the Python language. While I have had limited exposure to Python, I have rarely ever coded in Python, and never in the context of cloud-based software agents. Another facet complicating development of the project demo was that I needed to do extra background research to find solutions in Python for standard web development issues (such as AWS lambda deployment) that I was more familiar with in other web programming languages, such as Javascript and PHP. The net benefit of doing so, however, was an increased comfort level for working with Python.

V. Conclusion

The Conclusion step summarizes the conclusions from the preceding steps of the cycle (Praveena et al, 2025). Overall, my view is that the module and both assignments were successful in accomplishing their objective of broadening and refining my grasp of pragmatic software agent development principles and best practices.

VI. Action Plan

The Action Plan step of the Gibbs' Reflective Cycle identifies opportunities to enhance learning outcomes and to minimize learning challenges for future study (Praveena et al, 2025).

For future projects in a similar vein, I would place more emphasis at the design stage on limiting system components and features to only those truly necessary to build out a fully functional demo, or, in other words, a minimum viable product. The purpose of that approach would be to spend development time on only those system components that matter the most to accomplishing system outcomes and development deliverables, which would also likely provide some level of insight on how useful, or even necessary, less critical components might or might not be. Another way to express the same idea would be that I would focus more on an iterative approach to designing an agentic system, rather than a monolithic, "fully designed in advance" approach.

I would also allow more time to brush up on Python coding best practices, as I had little prior experience with Python coding, and found that part of the assignment to be quite time-consuming in comparison to programming languages that I am more familiar with (primarily Javascript, PHP, and Rust).

VII. References

Bullen, Matthew (2026). IA_PCOM7E January 2026 Group Design Proposal.

Bullen, Matthew (2026). IA_PCOM7E January 2026 Individual Development Project.

Praveena, K.S., Juslin, F., Patil, C.M. and Bhargavi, K. (2025). Using Gibb's Reflective Model Approach for Enhancing Project-Based Learning Among Students Through Reflective Assessment. *Journal of Engineering Education Transformations*, [online] 38(2), pp.148–155. doi:https://doi.org/10.16920/jeet/2025/v38is2/25018.

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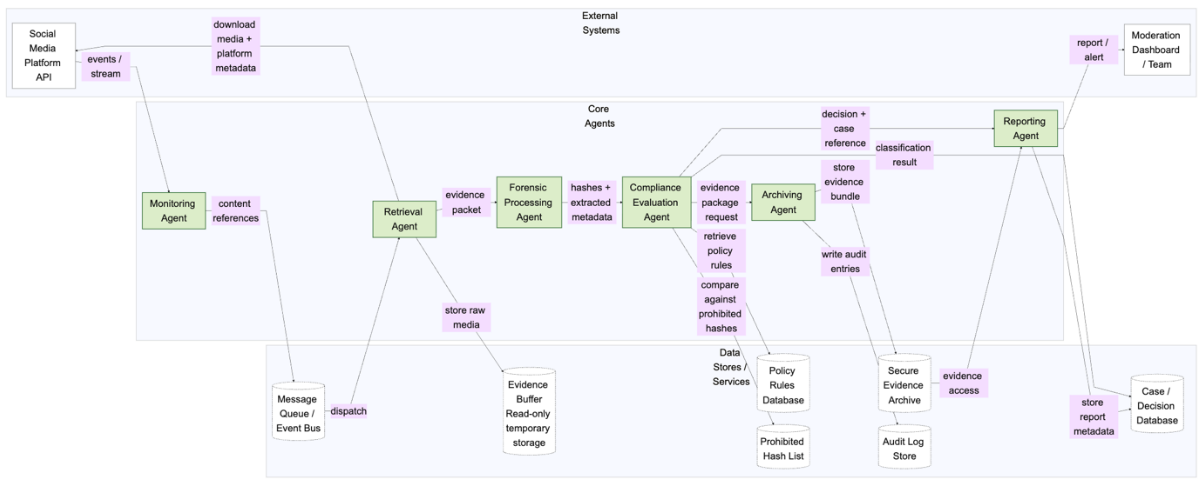
Appendix A: Architectural Diagram from the Group Design Proposal



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Appendix B: Implemented System Components in the Live Demo